SAEs, mappings, and autointerpretation

Supplement to the theoretical-analysis handoff

10 September 2026

What is included

This supplement adds the actual checkpoint locations, configuration files, saved mappings, feature descriptions, and interpretation examples. `sae_artifact_manifest.csv` records pinned download URLs, sizes, hashes, and exact package locations. Small files are included under `sae_assets/`; checkpoints and other files larger than 10 MB are linked, not embedded in the ZIP. No SAE training, map fitting, inference, or autointerpretation was run.

The theory report remains a dated results snapshot. This supplement adds assets; it does not refresh the China experiment's status or change the paper.

1. Choose a compatible pair of SAEs

All four dictionaries below concern Qwen2.5-7B-Instruct layer 19, with residual-state dimension 3,584. A context vector is the state at the final prompt token. A turn-averaged answer vector averages assistant-token states; it is **not** a per-token SAE or an average of separately encoded token features.

Family	Context SAE	Turn-averaged answer SAE	Use
Matched #2552 pair	32,768 features, BatchTopK, k=128	32,768 features, BatchTopK, k=128	Global/minimal-refusal dashboards and #2643 direct SAE maps
Earlier #2569/#2476 pair	65,536 features, Matryoshka BatchTopK, k=100	65,536 features, Matryoshka BatchTopK, k=100	Original theory battery and its feature-firing/magnitude regression

Checkpoint directories, each with its configuration:

- **32k context:** sae_ctx_rep, file `sae_weights.safetensors` with `cfg.json`.
- **32k answer:** sae_rep, file `sae_weights.safetensors` with `cfg.json`.
- **65k context:** sae_ctx, file `ae.pt` with `config.json`.
- **65k answer:** sae_c, file `sae_weights.safetensors` with `cfg.json`. Despite the folder name, this is the **answer** dictionary in the original theory analysis.

The 32k checkpoints are about 0.94 GB each; the 65k checkpoints are about 1.88 GB each. Preserve configurations, thresholds, normalization, and feature ordering. Do not pair a 32k dictionary with a 65k map or transfer descriptions by integer feature ID across dictionaries.

2. The mappings are different objects

Dense context-to-answer map

The frozen layer-19 ridge checkpoint maps residual states, not SAE codes. Its stored row-vector convention is

$$\hat{a} = ((c - x_\mu)/x_{sd})W + y_\mu.$$

The linear operator used by the geometry dashboards is $A = \mathrm{diag}(1/x_{sd})W$. For absolute-state predictions, keep the centering and output mean; for paired differences, the affine offset cancels. A read/write feature ranking based on this dense operator is not itself a learned SAE-to-SAE coefficient matrix.

Composed 32k SAE route

The original #2643 route is

$$\hat{z}_A = s \odot E_A(F(D_C(z_C))), \qquad z_C = E_C(c),$$

where E and D are the corresponding SAE encoder/decoder, F is the dense affine map, and s is the saved feature-wise calibration. The composition is not globally linear because it includes the answer SAE encoder.

Use the 32k pair, the dense ridge above, and feature_calibration.pt. The calibration is bundled. The manifest also includes the screen summary and separate refusal readouts; those readouts are not interchangeable with the SAE map.

Direct 32k context-code to answer-code maps

These are separately trained reduced-rank regressions:

$$\hat{z}_A = ((z_C - \mu_C)/s_C)B_1 B_2 + \mu_A.$$

Here B_1 and B_2 are payload fields `a` and `b`, and the means are `x_mean` and `y_mean`. The producer replaces every saved `x_std` entry below 10^{-8} with 1 before dividing; use that guarded standard deviation as s_C. This matters for inactive features. Both axes refer to their own 32k dictionaries, not shared feature indices. Predictions from this linear regression need not be nonnegative.

- Original rank-256 fits: `direct_rank256_seed2643.pt` and `direct_rank256_seed2644.pt`, plus training summaries and top-edge arrays.
- Tuned rank-1024 fits: `selected_seed2643.pt` and `selected_seed2644.pt` are validation-selected fits; `trainval_refit_seed2643.pt` and `trainval_refit_seed2644.pt` are the separately reported train-plus-validation refits. Selection metadata, final summary, edge arrays, and dashboard graph are included.

Keep both seeds and distinguish selected fits from refits. The graph's top edges are a selected view, not the entire matrix, and regression edges do not establish causal connections.

Older 65k feature regression: an explicit archive gap

The original leg-4 archive contains encoded inputs, target feature IDs, held-out predictions, and metrics. The checked producer returns predictions from its fit and does **not** serialize the fitted coefficient matrix. Thus this package supplies the evidence and input artifacts, but not an inference-ready checkpoint for that specific regression. The #2643 checkpoints are available alternatives for the **32k** pair, not replacements for this missing 65k fit. Nothing was refitted to fill the gap.

3. Autointerpretation and maximum-activation examples

There are three distinct description sources to consult:

1. **Original answer-SAE W1 descriptions:** raw W1 files. Both `judge_raw_w1.json` and `judge_raw_w1_syncreissue.json`, and their saved draw records, are bundled. For the 32k answer dictionary, use entries whose IDs match `w1-rep_ta-f<ID>`; do not treat every record in a mixed instrument as that dictionary's feature. The existing loader reads the main file followed by the reissue, with later valid descriptions taking precedence. These are historical outputs; no Claude or other model was invoked for this handoff.
2. **Context/answer edge-candidate autointerpretation:** autointerp.json and examples. These cover 64 context and 64 answer candidates, not all features. Answer descriptions reuse available W1 outputs; new descriptions were produced with Qwen2.5-7B-Instruct. Context examples came from the refusal panel rather than a broad context corpus. The saved protocol records those deviations.
3. **Later Codex interpretation and broad review:** the original handoff's `artifacts/codex_interpretations/` includes `codex_interpretation_results_final.json`, `codex_interpretation_report_final.md`, packets, and broad-review evidence. These are the later descriptions for the requested global and minimal-refusal mean/RMS extreme lists. Use them for that analysis, while retaining the original W1/edge labels for provenance. Coverage is selective, not a full-dictionary annotation.

The older 65k answer instrument's consumed descriptions are also bundled. They must not be attached to the 32k answer dictionary. No comparable complete 65k context-description inventory is claimed; the theory package retains its selected direction/SAE matches and examples.

4. Practical reuse

Start with the 32k pair and choose either the composed route or a direct fitted map. Read its configuration and selection metadata before loading weights. Join descriptions by **checkpoint identity, side, and feature ID**, not by feature ID alone. Missing descriptions mean unannotated features, not absent semantics. Distinguish saved maximum-activation examples from representative behavior across the full data distribution.

`sae_producers/` contains current #2643 producer and loader source snapshots, with hashes in `sae_producer_manifest.json`. In particular, inspect `issue2643_sae_map.py`, `issue2643_direct_edges.py`, and `issue2643_edge_autointerp.py` for the saved formats and description-merging rules. Use them alongside the project checkout and its dependencies; do not launch training or autointerpretation phases merely to open saved assets. These working-tree snapshots are not asserted to be byte-identical to every historical training script.

The supplement's CSV is the authoritative download list. Its small bundled files were checked against immutable Hub Git-blob or LFS hashes; large files were availability-checked but not downloaded or executed. The original artifact inventories remain unchanged, and `sae_asset_summary.json` records the additional coverage. This is an asset handoff, not a new evaluation or a claim of complete semantic interpretation.